BUSS386 Problem Set 7

Properties of Options: Bounds and Parity

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Problem 1 — Six determinants

Consider European and American calls and puts, all with the same strike and maturity on the same stock (which may pay dividends).

(a) Fill in the direction (↑, ↓, or "?" if the sign is ambiguous) in which each price moves when the factor *increases*:

Factor (increase ↑)	Eur. c	Eur. p	Am. C	Am. P
Spot S_0				
Strike K				
Volatility σ				
Dividends D				
Risk-free rate r				
Time to expiry T				

(b) Five of the six rows are identical across all four columns. Identify the row that is not, and explain why the European and American answers differ there.

(c) Explain why higher σ raises *both* c and p.

(d) Explain why an American put can be worth strictly more than the corresponding European put, but an American call on a non-dividend stock cannot.

Problem 2 — Put-call parity bounds for American options

A non-dividend-paying stock trades at $S_0 = \$100$. American calls and puts on it have $K = \$100$ and $T = 0.5$ yr. The risk-free rate is $r = 6\%$ c.c.

(a) State the American put-call parity bounds and compute the numerical range within which $C_0 - P_0$ must lie.

(b) The call trades at \$8.50 and the put at \$6.50. Is there an arbitrage?

(c) Now suppose the put trades at \$3.00 instead, with the call still at \$8.50. Show the arbitrage: list the trades, the cash flow today, and the cash flows at T in both states $S_T \geq K$ and $S_T < K$. What is the profit?

(d) In (c) the short call is American, so you may be assigned before T. Show that early assignment leaves you no worse off.

(e) Why is the lower bound $S_0 - K$ rather than the tighter $S_0 - Ke^{-rT}$ that applies to European options?

Problem 3 — Put-call parity check

SPX is at 5,800. A 3-month European call with $K = 5{,}800$ trades at \$140. A 3-month European put with the same K trades at \$80. The 3-month risk-free rate is $r = 4\%$ (c.c.); assume no dividends.

(a) State the put-call parity identity.

(b) Verify (or reject) parity for these quotes. Compute the discrepancy.

(c) If parity is violated by more than the bid-ask spread, sketch the arbitrage strategy (which side is rich, which is cheap, and which portfolio do you buy/sell?). *One short paragraph; no detailed cash flow table needed.*

Problem 4 — Lower-bound arbitrage on a European put

A European put on a non-dividend stock has $K = \$40$, $T = 0.5$ yr, $r = 5\%$ c.c. The stock is at $S_0 = \$37$ and the put is quoted at \$1.00.

(a) Compute the no-arbitrage lower bound on the put price.

(b) Show the arbitrage strategy in a 3-row cash-flow table (today, $S_T \geq K$, $S_T < K$).

(c) Compute the arbitrage profit today.

Problem 5 — Forward-style parity on KOSPI 200

KOSPI 200 spot $S_0 = 1{,}105$. A 6-month European call with $K = 1{,}105$ trades at 61.50 index points; the European put at the same K trades at 51.20. $r = 3\%$ c.c., dividend yield $q = 1.5\%$. Multiplier = ₩250,000 per point.

(a) Compute the 6-month forward price F_0 of the KOSPI 200.

(b) State the forward-style parity and verify it for the given quotes.

(c) If parity is violated, identify which side is rich and sketch the arbitrage trade.

(d) Compute the arbitrage profit per contract in ₩ (ignore bid-ask).

Problem 6 — Discrete-dividend parity

A 1-year European call on a stock with $K = \$50$ trades at \$6. The stock is at $S_0 = \$52$ and will pay a \$1.50 dividend in 6 months. $r = 4\%$ c.c.

(a) Compute the PV of dividends D.

(b) Use put-call parity to find the no-arbitrage price of a 1-year European put with the same strike.

(c) Without recomputing, by how much would the put price change if the dividend doubled to \$3.00?

Problem 7 — Early exercise of American options

(a) State the result for early exercise of an American call on a non-dividend-paying stock and prove it using put-call parity for European options.

(b) Give one concrete situation (state the parameter values you choose) in which early exercise of an American call *with* a discrete dividend would be optimal just before ex-dividend. Verify your numerical claim against the heuristic rule

$$D \;>\; K(1 - e^{-r(T-t_d)}) + p_{t_d}.$$

(c) Explain why early exercise of an American put can be optimal even with no dividends, in terms of the interest forgone vs the time value of waiting.